Does a Realistic Market Simulator Give Reliable Execution Decisions?

A Feasibility Audit, a Cross-Simulator Pilot, and a Controlled Reference-Market Test of ABIDES, MarS and a Hawkes Order-Book Simulator

Gaurav Chattree*

October 2026

Abstract

Limit-order-book simulators are validated mostly by how realistic their output looks, and used to decide how to execute large orders. We examine whether the first implies the second for three simulator families: ABIDES (agent-based), Microsoft MarS (a generative order-flow model) and a marked multivariate Hawkes order-book simulator. A feasibility audit comes first. We had no licensed order-level data, so realism against real markets was not measured; MarS's released models run on a laptop and respond to injected orders, but cannot be refitted and were unstable, failing in 7 of 12 and 4 of 12 unperturbed runs for its two models. All simulators were then put behind one execution interface with twelve candidate policies and identical cost accounting. In a cross-simulator pilot, policy rankings agreed weakly within a simulator family (mean rank correlation 0.34) and not across families (-0.05). Because disagreement cannot identify who is right, we built four reference markets with known rules, gave each simulator only unperturbed data from them, and measured the cost, in the reference market, of the policy each fitted simulator recommended. Realism did not predict decision quality. The Hawkes fit matched the reference markets' stylised facts best in three of four cases, yet in the two markets where policies differ materially it recommended immediate execution of the larger order, 21 to 22 basis points worse than the best candidate, while a less realistic ABIDES fit was within 2 to 5. Three Hawkes variants with the same realism score recommended different policies. MarS, conditioned on reference order flow, could not be stabilised in any reference market. Averaged over settings, always choosing TWAP (regret 1.2 basis points) beat ABIDES (2.4), a random candidate (4.1) and Hawkes (6.7). The results concern twelve policies in four synthetic markets and are not evidence about real markets.

1 Introduction

Limit-order-book simulators are judged mainly by how their output looks: spreads, volatility, order-flow statistics, and the other stylised facts collected in realism benchmarks (Vyetrenko et al., 2020; Nagy et al., 2025). They are used for something else. A trader who must execute a large order wants the simulator to say what a given execution policy will cost and which policy to prefer. Whether looking right implies deciding right is an empirical question, and it cannot be answered from historical data alone, because history never shows what would have happened under a different execution strategy.

This paper keeps three claims apart.

*Experiments, code and manuscript were produced with the assistance of an AI coding agent (Claude Code). Every number is generated by the scripts in the accompanying repository. Co-authors to be added.

1. **Observational realism.** How closely does a simulator's unperturbed market resemble a target market?
2. **Cross-simulator robustness.** Does a policy chosen in one simulator remain competitive in another?
3. **Controlled decision validity.** When the data-generating market is known, does a simulator fitted to its data recommend a policy that does well in it?

Agreement between simulators (claim 2) says nothing about correctness. Only claim 3 has a ground truth, and only inside the synthetic market that supplies it.

We study three simulator families: ABIDES (Byrd et al., 2019; Amrouni et al., 2021), an agent-based simulator; MarS (Li et al., 2025), a generative order-flow model; and a multivariate marked Hawkes order-book simulator (Bacry et al., 2015) written for this study. The work proceeds in the order a careful study must: a feasibility audit with explicit gates (Section 3), a common harness (Section 4), a cross-simulator pilot (Section 5), and a controlled experiment in which simulators are fitted to markets whose rules we wrote (Section 6).

What was and was not possible. We had no licensed order-level market data, so the realism of each simulator against held-out historical data was not measured, and no simulator was calibrated to a real instrument. MarS's released models run on a laptop and respond to injected orders, but they cannot be refitted (no training code is released), were trained on Chinese equities, and proved unstable in our hands; MarS is therefore a separate-domain case study in the pilot and could not be evaluated in the controlled experiment. Section 8 lists every limitation.

2 Related work

ABIDES (Byrd et al., 2019) simulates an exchange and a population of message-passing agents; its behaviour comes from the configured population. MarS (Li et al., 2025) trains an autoregressive model on order-level data and generates orders one at a time, conditioned on recent orders and the book, inside its own matching engine; its authors demonstrate forecasting, stylised facts and a market-impact study. TRADES (Berti et al., 2025) and M3 (Zhang et al., 2026) are other learned generators. Hawkes processes are the standard statistical model of clustered order flow (Bacry et al., 2015); queue-reactive models make intensities depend on the state of the book instead of on past events (Huang et al., 2015; Cont et al., 2010).

Realism of simulated order books is assessed with stylised facts (Vyetrenko et al., 2020) and with distributional and price-response scores (Nagy et al., 2025). Balch et al. (2019) argue that interactive simulation is needed to evaluate strategies because replay cannot react. Optimal execution theory supplies the policies we test (Almgren and Chriss, 2001). The question of whether simulator realism predicts the quality of the decisions a simulator supports is, to our knowledge, less studied; we make no claim of priority, having run targeted searches only.

3 Feasibility audit

Each candidate had to pass five gates before any comparison: (A) runnable from a pinned commit with a reproducible seeded episode; (B) interactive, meaning an external order schedule can trade and the market responds; (C) comparable in market conventions and regime; (D) stable across seeds; (E) reproducible in data rights and compute. Table 1 gives the outcome; the full evidence is in `simrel/reports/feasibility_matrix.md`.

Table 1: Feasibility gates. [a]After setting the price level so the tick matches. [b]For the configurations used; others were unstable in an earlier pilot. [c]On CPU, outside the supported Docker/CUDA set-up.

Simulator	A Runnable	B Interactive	C Comparable	D Stable	E Reproducible
ABIDES	Pass	Pass	Pass[a]	Pass[b]	Pass
MarS (2M, 10M)	Pass[c]	Pass	**Fail**	**Conditional**	Pass
Hawkes order book	Pass	Pass	Pass	Pass	Pass

ABIDES. We use `abides-jpmc-public` at commit `f9cbe51`. It runs on current Python once one obsolete dependency is replaced by a small stand-in, and reproduces results bit-for-bit across Python versions. An earlier pilot of ours found two of six configurations unstable without any intervention; here we use only the reference configuration and one with more market making, and, in the controlled experiment, configurations at a lower price level so that the tick is the same fraction of the price as in the reference markets.

MarS. The repository (commit `f04dd87`) releases order models of 2M, 5M and 10M parameters under the MIT licence; the larger models reported in the paper are not released. The supported set-up is Linux, Docker, CUDA and a Ray Serve deployment. The released models are small enough to run in-process on a CPU (14 to 24 ms per generated order), so we replaced the serving client with a local one and left the repository untouched. We checked the local inference against the official validation samples: next-order cross-entropy is 8.17, 7.93 and 7.75 for the three sizes against 10.80 for a uniform guess, falling with model size as in the published scaling curve.

MarS passes gates A, B and E. It fails C for any real market: it was trained on Chinese equities, ships no training code, and its demonstrations start from a synthetic noise-agent state. On D it is conditional at best. Reproducing the official market-impact example, the unperturbed mid price moved by 463 bps on average within five minutes, and by 2,453 bps with the repository's own TWAP agent buying 30% of volume. Over thirty unperturbed minutes, five-minute moves above 200 bps occurred in 25% of windows during the first fifteen minutes after hand-over from the noise agent (10M model), and in none of the windows after that. We therefore let MarS run for twenty minutes before any task. Even so, rollouts fail: some empty their own order book, after which the engine raises an error, and some enter a burst of order generation that does not end within a wall-clock limit. Section 5 reports the failure rates.

Hawkes. A Hawkes event generator alone cannot answer execution questions, because prices must respond to orders. Ours has six mutually exciting event types (limit, market and cancel, on each side) with exponential kernels and size and price-level marks, applied to a price–time-priority book. Cancellation intensity scales with resting depth and market-order intensity falls when the spread is wide. An injected market order hits the same book and enters the intensities as an event weighted by its size, so later flow reacts to it. Events are generated by thinning (Ogata, 1981) candidate streams that do not depend on the market state, which keeps an intervention and its unperturbed twin on the same random path wherever their states agree.

Consequence. Following the stop rule fixed in advance, we did not run a three-way comparison on harmonised real data. We ran the pilot on each simulator's own domain, with MarS labelled as a separate domain, and the controlled experiment, in which the common domain is a market we generate.

4 Common harness

Task. Sell a parent order over ten minutes in twenty decisions thirty seconds apart, then observe for two more minutes. The parent size is 5% and 15% of the world's own mean background volume over the window.

Policy interface. Every policy is a function

$$\mathit{quantity} = \pi(\mathit{observation}_t,\ \mathit{remaining\ inventory},\ \mathit{time\ remaining},\ \mathit{state}),$$

where the observation holds best quotes, depth at one and five levels, the previous slice's traded and signed volume and mid change, and the arrival mid: nothing later than the decision time. The returned quantity is sent as one market order. The harness, not the policy, clips it to the remaining inventory, rounds it to the lot, and sends everything left at the last decision, so all policies complete the same order and any forced liquidation is in their cost. Unit tests check the sign conventions, these constraints, and that no policy's decisions up to a time change when later observations are altered.

Policies. Twelve candidates in eight families: TWAP (P0); immediate execution limited to the visible five-level depth per decision (P1); two front-loaded schedules (P2) and one back-loaded schedule (P3); participation-of-volume at one and two times the rate that would complete on time (P4); a liquidity-aware rule that speeds up when the spread is tight and depth ample relative to what the episode has shown (P5); a signal-aware rule tilted by the last slice's order-flow imbalance and return, in a momentum and a reversal variant (P6); and an Almgren–Chriss feedback controller whose urgency is estimated online from realised volatility and the spread, at two risk aversions (P7). The optional MFG-derived policy was not included.

Cost. For a sell, implementation shortfall is $10^4\,(m_0 - \bar{p})/m_0$ with m_0 the arrival mid and $\bar{p}$ the volume-weighted fill price; shares the market could not absorb are charged at the worst slice price. Positive is worse for buys and sells alike. We also report the *impact cost*, the same fills measured against the unperturbed twin's mid at each decision.

Twins. Every intervention is compared with an unperturbed run from the same state. In MarS, the Hawkes simulator and the reference worlds, the whole environment is copied once per policy at the task start, so all policies and the twin share an identical history. ABIDES cannot be copied; its twin is a same-seed rerun that is identical until the first child order.

5 Pilot: cross-simulator robustness

The pilot runs the twelve candidates at both sizes in six worlds: two ABIDES configurations (A1, the RMSC04 reference; A5, with more market making), two hand-set Hawkes regimes (H1 liquid, H2 thin), and MarS with the 10M and 2M models (M10, M2). Each world keeps its own conventions, so this is a comparison across domains, not a like-for-like one. In total 5,304 policy rollouts were attempted and 5,060 are used.

Table 2: Stability audit of the pilot. Volume pass: unperturbed runs used to size the parent order. Main pass: a seed fails if its burn-in or its unperturbed twin fails; it is screened out if the mid moved more than 200 bps in the five minutes before the task. Last three columns describe the unperturbed twin over the ten-minute task in the seeds used: standard deviation of the mid move, share of seeds moving more than 3%, and mean quoted spread.

World	Family	Volume pass		Main pass (seeds)				Move s.d.	>3%	Spread
		run	failed	run	failed	screened	used	(bps)		(bps)
A1	abides	20	0	40	0	0	40	5	0%	0.2
A5	abides	20	0	40	0	0	40	5	0%	0.1
H1	hawkes	20	0	60	0	0	60	52	0%	11.7
H2	hawkes	20	0	60	0	3	57	153	5%	20.2
M10	mars	12	7	10	4	1	5	63	0%	14.6
M2	mars	12	4	11	1	0	10	277	10%	21.7

5.1 Stability comes first

Table 2 is the audit that precedes any cost number. ABIDES and the Hawkes simulator completed every run. MarS did not. In the volume pass, 7 of 12 and 4 of 12 unperturbed rollouts (10M and 2M) failed: the burn-in or the rollout exceeded its time limit in a burst of order generation, the book emptied and the engine raised an error, or an internal assertion fired. In the main pass, 5 of 10 and 1 of 11 seeds were lost to failure or to the pre-task screen, and of the policy rollouts on the remaining seeds a further 15 of 120 (10M) and 13 of 240 (2M) failed. Because of the time limits, some of these are slow rollouts on a loaded machine, not crashes; all of them are runs that a practitioner would not obtain a number from. The surviving sample is therefore selected on stability, and it is small: five seeds for the 10M model.

Among surviving seeds the 2M model's unperturbed mid still moves by more than 3% in ten minutes in 10% of cases, with a standard deviation of 277 bps, five times that of the liquid Hawkes regime and fifty times that of ABIDES.

5.2 Costs and what each world recommends

Table 3 gives shortfall at 15% of volume (5% is in Appendix D). Three patterns matter.

ABIDES barely distinguishes the candidates. All costs are below 4 bps and most intervals overlap. Only the immediate order, the fast front-loaded schedule and the risk-aware controller are visibly dearer.

The Hawkes regimes price impact an order of magnitude higher, at 35 to 80 bps, and disagree with each other about adaptive rules: the liquidity-aware rule is among the best in the liquid regime and among the worst in the thin one.

MarS inverts the ordering, violently. With the 10M model, TWAP costs 257 bps and the immediate order 39 bps at 15% of volume; at 5% the gap is 905 against 20 bps. A persistent seller is met by a market that keeps falling (Figure 1), which is the behaviour already visible in the official example. With five usable seeds and intervals of hundreds of basis points these are anecdotes, but they are consistent in sign across sizes for the 10M model. The 2M model does not show the same ordering.

Table 3: Pilot: implementation shortfall in bps (mean ± 95% half-width) for a parent order of 15% of background volume. Bold: lowest mean in the column. Costs are not comparable across columns: the worlds have different price levels and tick sizes.

Policy	A1	A5	H1	H2	M10	M2
P0 TWAP	0.4 ±1.0	0.3 ±0.9	39.0 ±10.2	58.0 ±28.2	257.2 ±339.8	28.8 ±28.4
P1 Immediate	1.7 ±0.5	1.1 ±0.3	43.8 ±5.9	57.4 ±16.5	**38.9 ±18.7**	34.5 ±25.1
P2 Front-loaded (b=2)	0.5 ±0.7	0.5 ±0.6	45.9 ±9.5	50.2 ±20.4	778.6 ±922.4	14.0 ±26.3
P2 Front-loaded (b=4)	1.4 ±1.3	0.7 ±0.4	44.9 ±9.2	54.1 ±18.0	79.7 ±45.9	3.5 ±17.9
P3 Back-loaded	0.5 ±1.2	0.3 ±1.1	38.5 ±11.9	69.9 ±33.0	97.7 ±85.8	**0.0 ±38.0**
P4 POV (1×)	0.4 ±0.8	0.3 ±0.8	46.7 ±11.9	68.2 ±27.7	564.7 ±348.2	5.5 ±12.1
P4 POV (2×)	1.1 ±0.7	0.8 ±0.5	45.1 ±6.7	50.2 ±21.4	411.7 ±411.7	22.3 ±27.0
P5 Liquidity-aware	0.5 ±0.9	0.3 ±0.9	36.8 ±12.8	78.0 ±27.5	209.8 ±201.5	7.5 ±19.1
P6 Signal (momentum)	0.7 ±1.0	0.9 ±1.0	40.0 ±12.0	80.3 ±28.1	42.0 ±271.4	21.7 ±28.1
P6 Signal (reversal)	**0.3 ±0.9**	**-0.1 ±0.9**	40.5 ±10.5	63.7 ±26.4	346.6 ±457.9	20.7 ±30.8
P7 Risk-aware (0.5)	3.4 ±2.2	1.1 ±0.4	38.0 ±5.9	37.8 ±18.7	258.1 ±347.1	29.0 ±33.4
P7 Risk-aware (2)	2.7 ±0.9	1.4 ±0.4	**36.3 ±5.3**	**35.4 ±19.3**	235.5 ±278.3	15.6 ±11.9
Seeds	40	40	60	57	5	10
Half-spread (bps)	0.1	0.1	5.8	10.1	7.3	10.8

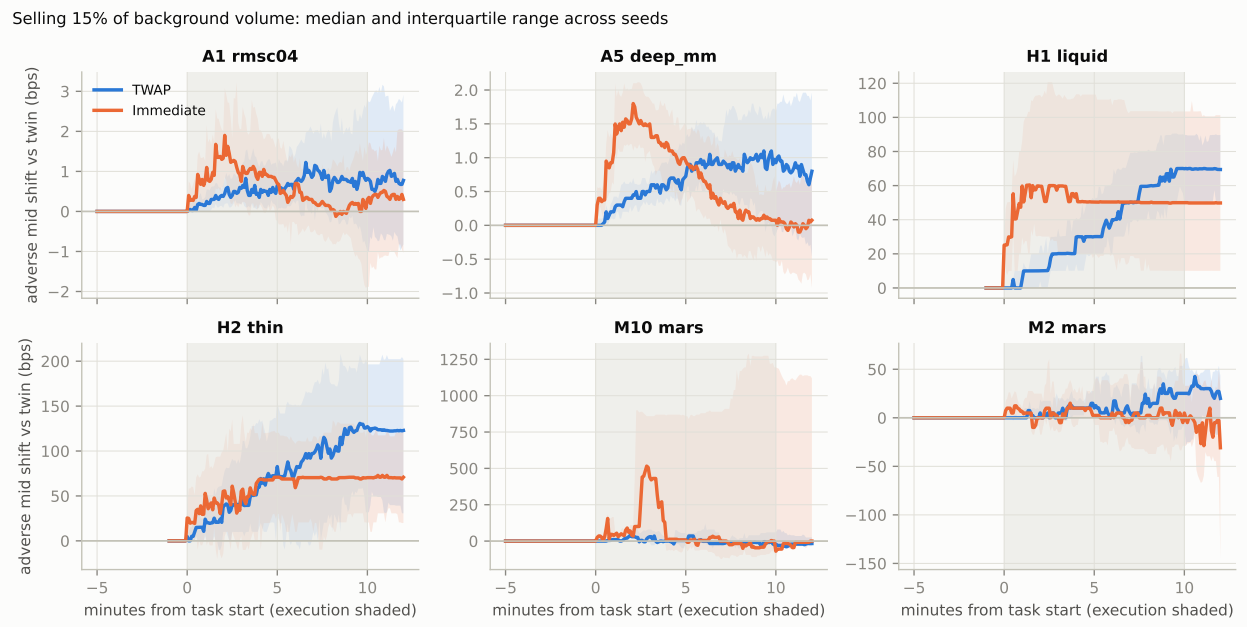


Figure 1: Intervention against no intervention: adverse shift of the mid price relative to the unperturbed twin, for TWAP and immediate execution of 15% of background volume. Lines are medians across seeds, bands the interquartile range. Vertical scales differ by three orders of magnitude.

5.3 Transfer between simulators

Selecting each world's cheapest candidate on half its seeds and evaluating it on the other half of every world's seeds gives the cross-simulator matrix (Table 11 in the appendix, Figure 2). Policy rankings agree weakly within a family (mean Spearman correlation 0.34) and not at all across families (-0.05). Of the 60 off-diagonal transfers, 22 carry a regret whose interval lies above zero. The largest regrets, tens of half-spreads, arise when a policy chosen elsewhere is evaluated in MarS, or when a policy tuned to the Hawkes regimes is evaluated in ABIDES, whose half-spread is a tenth of a basis point.

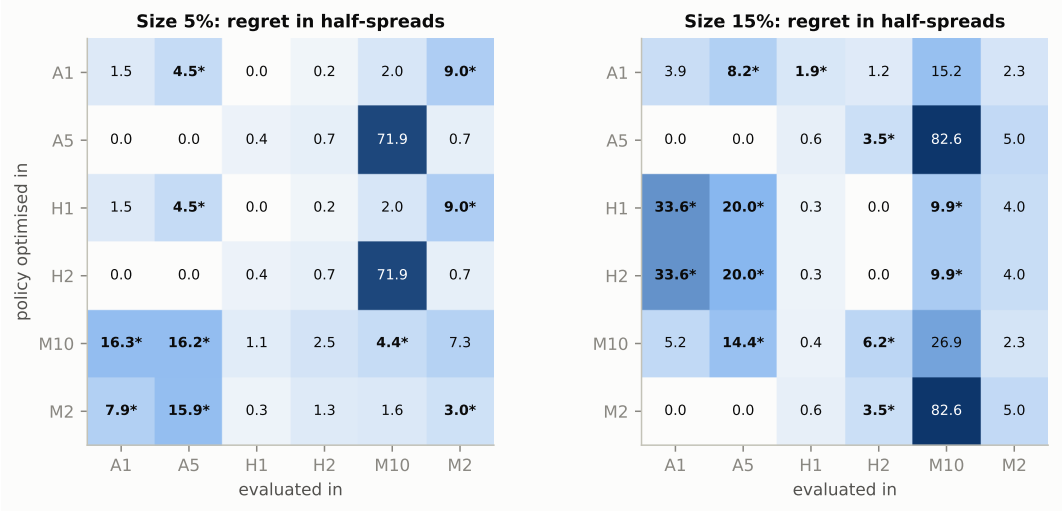


Figure 2: Cross-simulator regret in units of the evaluation world's half-spread. An asterisk marks a paired 95% interval above zero. Colour is on a square-root scale.

This establishes that the simulators do not agree, and nothing more. The worlds differ in price level, tick, liquidity and calibration, and none of them is tied to a real market. Disagreement cannot say which, if any, is right. That needs a ground truth.

6 Controlled decision validity

6.1 Design

We wrote 4 reference worlds from two mechanisms that are deliberately unlike any of the candidates (Table 4, Appendix A): a queue-reactive book, where order flow depends only on the current queues, and a mix of behavioural traders (noise, fundamentalists, trend followers, and liquidity providers who withdraw from one-sided flow) with a hidden regime. Each candidate simulator receives 6 unperturbed 30-minute episodes from a reference world, as event logs and quotes, and nothing else. It is fitted to that data, then asked which of the twelve candidates it prefers at each size. Separately, every candidate policy is run in the reference world itself on 300 seeds; half select the best-in-class policy and the other half measure regret.

The *reference regret* of a simulator is the cost, in the reference world, of the policy it recommends minus the cost of the best candidate there. It is zero for a simulator that picks the best candidate and is bounded by the worst. It is a statement about these twelve policies in these four synthetic worlds.

Table 4: Reference worlds.

World	Mechanism and regime
R1a	Queue-reactive, deep and resilient
R1b	Queue-reactive, thin, flow chases imbalance
R2a	Agent mix, strong fundamental anchor
R2b	Agent mix, trend followers dominate

6.2 The truth being recovered

Table 5: Reference worlds: the truth each simulator is judged against. Best: lowest-cost candidate on half the reference seeds; its cost and all regrets (bps) are measured on the other half. Sig. worse: candidates whose paired 95% interval lies above the best. Raw and adjusted s.d.: per-seed standard deviation of shortfall before and after the control-variate adjustment.

World	Size	Best	Cost	TWAP	Immediate	Worst candidate	Sig. worse	Raw s.d.	Adj. s.d.
R1a	5%	POV 1×	5.1	+0.0	+0.0	Front b=4 (+0.1)	0 / 11	2	1
R1a	15%	Momentum	5.7	-0.3	+5.2	Immediate (+5.2)	1 / 11	3	2
R1b	5%	Liquidity	15.0	+0.1	+2.3	Immediate (+2.3)	0 / 11	26	16
R1b	15%	Liquidity	35.2	-2.5	+12.1	Immediate (+12.1)	1 / 11	30	21
R2a	5%	Reversal	3.0	+3.0	+12.4	Immediate (+12.4)	11 / 11	33	16
R2a	15%	Reversal	7.6	+2.6	+21.1	Risk 2 (+22.9)	11 / 11	33	17
R2b	5%	Reversal	5.0	+3.1	+11.4	Risk 2 (+14.8)	11 / 11	45	14
R2b	15%	Reversal	11.2	+3.7	+22.0	Risk 2 (+24.5)	11 / 11	44	14

Table 5 shows that the worlds are not equally demanding. In the queue-reactive worlds the candidates are nearly interchangeable: only the immediate order is significantly worse than the rest, and only at the larger size. A recommendation can hardly go wrong there. The agent-mix worlds are discriminating. The reversal variant of the signal-aware rule is best in both regimes at both sizes, every other candidate is significantly worse, and the immediate order and the more risk-averse controller cost 21 to 25 bps more at 15% of volume. The reason is built into the mechanism: after an adverse move, fundamentalists push the price back, so waiting is rewarded.

6.3 The fitted simulators

Hawkes. The fit attributes much of the agent-mix worlds' order clustering to self-excitation: the estimated branching ratio is 0.34 and 0.47 in the queue-reactive worlds and 0.75 and 0.84 in the agent-mix worlds. Its unperturbed output is close to the reference on the realism score in three worlds (0.21, 0.29 and 0.60) and within about two units in the fourth.

ABIDES. The grid search chose 180 value traders in three worlds and 102 in the fourth, with market makers in two. It matches the reference less well (1.16, 0.79, 0.58 and, in the deep queue-reactive world, 10.6, where ABIDES is far too volatile), and its traded volume is 28% to 96% of the reference's.

MarS. Conditioned on replayed reference order flow, MarS did not produce a usable market in any world. Between 17% and 33% of its unperturbed runs failed, and of the rest 40% to 50% moved by more than 3% within ten minutes, with traded volume two to four times the reference's. Under the screen in Table 6 it was not asked for a recommendation. This is a statement about a

Table 6: Fitted simulators in each reference world. Runs failed and moves above 3%: unperturbed runs of the fitted simulator. A simulator is evaluated only if at most half fail and at most a quarter of the rest move more than 3% in ten minutes. Realism error: mean scaled distance of six stylised facts from held-out reference data (lower is better; about 1 is within the reference world's own run-to-run variation). Volume ratio: simulator volume over reference volume.

World	Simulator	Runs failed	Moves >3%	Evaluated	Realism error	Volume ratio
R1a	Hawkes	0%	0%	yes	1.72	0.99
R1a	ABIDES	0%	0%	yes	10.63	0.96
R1a	MarS	33%	50%	**no**	–	3.67
R1b	Hawkes	0%	0%	yes	0.21	0.97
R1b	ABIDES	0%	0%	yes	1.16	0.54
R1b	MarS	17%	40%	**no**	–	1.88
R2a	Hawkes	0%	5%	yes	0.29	0.82
R2a	ABIDES	0%	0%	yes	0.79	0.40
R2a	MarS	33%	50%	**no**	–	3.26
R2b	Hawkes	0%	0%	yes	0.60	1.21
R2b	ABIDES	0%	8%	yes	0.58	0.28
R2b	MarS	33%	50%	**no**	–	1.79

pretrained model placed, without fitting, into a market unlike its training data; it is not a test of the MarS approach.

6.4 Reference regret

Table 7: Reference regret of each fitted simulator's recommendation (bps; †interval above zero). Tied: number of candidates the simulator cannot distinguish from its favourite; Worst tied: the largest reference regret among them. Rank ρ: Spearman correlation between the simulator's ranking of the twelve candidates and the reference world's.

World	Size	Simulator	Recommends	Ref. best	Regret	95% CI	Tied	Worst tied	Rank ρ	Realism
R1a	5%	Hawkes	Immediate	POV 1×	+0.0	[-0.2, 0.3]	12	+0.1	0.13	1.72
R1a	5%	ABIDES	Momentum	POV 1×	+0.1	[-0.1, 0.3]	4	+0.1	0.02	10.63
R1a	15%	Hawkes	Liquidity	Momentum	-0.1	[-0.3, 0.0]	5	+0.0	0.50	1.72
R1a	15%	ABIDES	Momentum	Momentum	+0.0	[0.0, 0.0]	2	+0.0	0.55	10.63
R1b	5%	Hawkes	Back	Liquidity	-2.5	[-5.8, 0.8]	12	+2.3	0.71	0.21
R1b	5%	ABIDES	Momentum	Liquidity	+0.8	[-2.7, 4.5]	2	+0.8	0.10	1.16
R1b	15%	Hawkes	Risk 2	Liquidity	-2.2	[-5.7, 1.5]	6	+1.8	0.59	0.21
R1b	15%	ABIDES	Liquidity	Liquidity	+0.0	[0.0, 0.0]	2	+0.0	0.60	1.16
R2a	5%	Hawkes	Liquidity	Reversal	+4.0†	[2.7, 5.5]	3	+4.0	0.66	0.29
R2a	5%	ABIDES	Momentum	Reversal	+5.3†	[3.8, 7.0]	2	+5.3	0.43	0.79
R2a	15%	Hawkes	Immediate	Reversal	+21.1†	[17.5, 25.0]	7	+22.9	0.04	0.29
R2a	15%	ABIDES	Liquidity	Reversal	+1.8†	[0.7, 2.9]	2	+3.8	0.69	0.79
R2b	5%	Hawkes	Immediate	Reversal	+11.4†	[8.5, 14.2]	4	+11.4	-0.13	0.60
R2b	5%	ABIDES	Liquidity	Reversal	+6.1†	[3.6, 9.1]	8	+9.0	0.65	0.58
R2b	15%	Hawkes	Immediate	Reversal	+22.0†	[18.0, 26.7]	4	+24.5	-0.84	0.60
R2b	15%	ABIDES	Momentum	Reversal	+4.9†	[2.9, 7.0]	3	+4.9	0.85	0.58

Table 7 and Figure 3 give the main result, and Table 8 summarises it.

Neither fitted simulator beats the simplest rule. Averaged over the eight settings, always choosing TWAP has a regret of 1.2 bps. The ABIDES fits average 2.4 bps, a uniformly random candidate

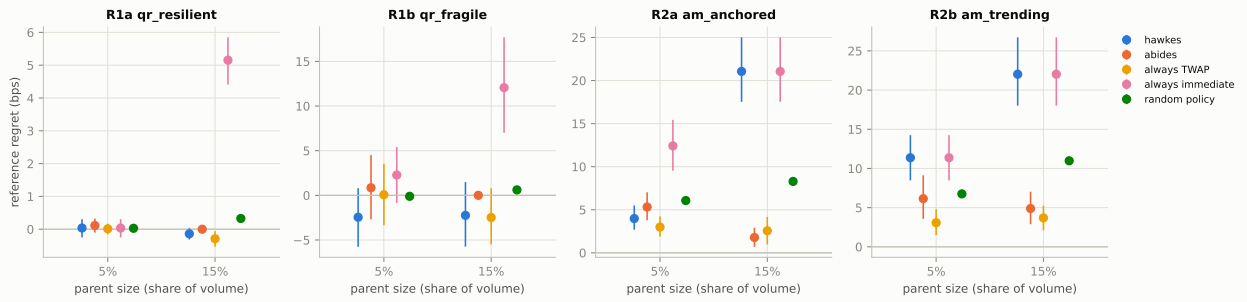


Figure 3: Reference regret of each recommender with 95% intervals. "Random policy" is the mean regret over the twelve candidates.

Table 8: Reference regret by recommender, over all reference worlds and sizes in which it was evaluated. The last four rows are simulator-free baselines.

Recommender	Settings	Mean regret (bps)	Mean (half-spreads)	Max (half-spreads)	Sig. worse than best
Hawkes	8	6.7	1.10	3.60	4
Hawkes (linear)	8	4.9	0.79	3.99	4
Hawkes (sqrt)	8	4.9	0.81	3.60	3
ABIDES	8	2.4	0.40	1.00	4
Always TWAP	8	1.2	0.19	0.60	4
Always immediate	8	10.8	1.87	3.60	6
Random candidate	8	4.1	0.69	1.79	0
Worst candidate	8	11.8	2.03	3.99	0

4.1 bps, the Hawkes fits 6.7 bps, and always choosing the immediate order 10.8 bps, against 11.8 bps for the worst candidate in each setting.

The simulator with the better realism score made the worst decisions. In the anchored agent-mix world the Hawkes fit has a realism error of 0.29 against 0.79 for ABIDES. At 15% of volume it recommends the immediate order, at a regret of 21.1 bps (interval 17.5 to 25.0), close to the worst available; ABIDES recommends the liquidity-aware rule, at 1.8 bps. In the trending world the two realism scores are within 0.03 of each other, and Hawkes again recommends the immediate order at both sizes (11.4 and 22.0 bps) while ABIDES is at 6.2 and 4.9 bps. Of the eight head-to-head comparisons between the two simulators' recommendations, four are statistically distinguishable: one small one favours Hawkes (1.3 bps) and three favour ABIDES (5.2, 17.1 and 19.3 bps).

Realism did not predict regret. Across the 16 simulator–world–size points the Spearman correlation between realism error and regret is -0.20 (permutation $p = 0.74$, shuffling realism scores among simulators within a world). With two simulators and four worlds this is descriptive, and the sign of a correlation this weak carries no information. The informative fact is the counterexample above.

Cost levels are not predicted either. The Hawkes fits overstate costs by 53 and 71 bps on average in the two agent-mix worlds at the larger size; the ABIDES fits overstate them by 8 and 67 bps. More telling is the sign of one comparison. In both agent-mix worlds the immediate order truly costs about 18 bps more than TWAP at 15% of volume. ABIDES predicts +10 and +54 bps. Hawkes predicts −9 and −50 bps: it believes splitting the order is the expensive choice.

6.5 Why the Hawkes fit fails, and an ablation

A Hawkes process explains clustered flow by letting events trigger further events. Fitted to the agent-mix worlds it finds a branching ratio near 0.8, so in its world each child order of a sliced schedule sets off its own cascade of selling and cancellations, and twenty child orders set off twenty. In the reference world the clustering has a different cause, traders reacting to prices, and part of that reaction pushes the price back. Both accounts fit the unperturbed data. They disagree about what happens when someone intervenes, which is the only thing an execution study asks.

The same point can be made inside the Hawkes family. How strongly a market order excites later flow as a function of its size is an assumption of the simulator: unperturbed data contain few orders of the size of a parent order, so they barely constrain it. Table 9 fits three versions (excitation linear in size up to ten mean order sizes; linear without an effective cap; square-root) to the same data.

Table 9: Three Hawkes simulators that differ only in how strongly a market order excites later flow as a function of its size, each fitted to the same reference data. Entries: recommended policy and its reference regret in bps (†interval above zero).

World	Size	Capped linear	Linear	Square-root
R1a	5%	Immediate (+0.0)	Immediate (+0.0)	Immediate (+0.0)
R1a	15%	Liquidity (-0.1)	Liquidity (-0.1)	Liquidity (-0.1)
R1b	5%	Back (-2.5)	Back (-2.5)	Reversal (-2.2)
R1b	15%	Risk 2 (-2.2)	Liquidity (+0.0)	Risk 2 (-2.2)
R2a	5%	Liquidity (+4.0†)	Liquidity (+4.0†)	Reversal (+0.0)
R2a	15%	Immediate (+21.1†)	Liquidity (+1.8†)	Immediate (+21.1†)
R2b	5%	Immediate (+11.4†)	Immediate (+11.4†)	Immediate (+11.4†)
R2b	15%	Immediate (+22.0†)	Risk 2 (+24.5†)	Front b=4 (+11.6†)
Mean realism error		0.71	0.71	0.76

Their realism scores are almost identical. Their recommendations are not: in the anchored world at 15% of volume the regret is 21.1, 1.8 and 21.1 bps; in the trending world it is 22.0, 24.5 and 11.6 bps. A choice that the realism score cannot see moves the decision from near-best to near-worst. None of the three does better on average than a random candidate.

7 Discussion

What the evidence supports. In this controlled setting, observational realism was not a guide to decision validity. A simulator that reproduced the reference market's stylised facts well recommended one of the worst policies, a less realistic one did better, and three simulators with the same realism score recommended different policies. The mechanism is not exotic: two models can fit the same unperturbed data and differ in what they imply about interventions, and a realism score computed on unperturbed data cannot separate them.

What it does not support. It does not show that agent-based simulators give better decisions than statistical ones. ABIDES did better than Hawkes here, in two reference worlds built from behavioural agents, which resemble ABIDES more than they resemble a Hawkes process; with other reference worlds the order could reverse. It does not show that realism is irrelevant, only that the score we used did not predict regret in sixteen points. And it says nothing about real markets.

The baseline result. That TWAP beat both fitted simulators is partly a property of our reference worlds, in which the cost surface is flat near TWAP. It is still a useful discipline: a simulator-derived recommendation should be reported against simulator-free baselines, because a simulator can do worse than not using one.

Stability is a gate, not a detail. MarS never reached the starting line in the controlled experiment and lost half its seeds in the pilot. We do not conclude that generative order-flow models are unfit for execution studies. We conclude that the released small models, started from the states available to us and run for tens of minutes, do not hold a market steady, and that their sensitivity to persistent flow, in which a patient seller moves the price by percent, should be checked before any cost estimate from them is used.

8 Limitations

- **No real data.** Realism against held-out historical data (the planned first experiment) was not measured, and no simulator was calibrated to a real instrument. All regret is inside simulated markets.
- **The reference worlds are ours.** Two mechanisms, two regimes each. "Best" means best of twelve candidates. Two of the four worlds barely discriminate between policies.
- **Unequal fits.** Hawkes shares the reference worlds' matching engine, lot and event vocabulary and is fitted by maximum likelihood on their event logs; ABIDES uses its own exchange and a three-parameter grid search; MarS could not be fitted at all.
- **A thin realism score.** Six statistics of a 5-second quote grid. It ignores order sizes, event mix, depth and inter-arrival times. A richer score, in particular one that includes responses to large natural orders, might have flagged the Hawkes fit. We did not test that.
- **MarS.** Small released models only; run on CPU outside the supported environment; trained on another market; initialised from synthetic states; child orders capped at visible depth; wall-clock limits contribute to its failure counts; surviving seeds are selected on stability and few.
- **Pilot comparability.** Three matching engines, different price levels and ticks. Cross-simulator regret in the pilot is exploratory.
- **Policies.** Market orders only; no limit orders, fees or latency; the optional MFG-derived policy was not included.
- **Selection noise.** A recommendation is the candidate with the lowest mean cost over a simulator's seeds (100 for Hawkes, 30 for ABIDES). Where candidates are tied in the simulator, the choice is partly chance; Table 7 reports the size of each tied set and the worst regret within it.
- **Post-hoc choices.** The stability screen for fitted simulators, the floor in the realism score, the ABIDES grid and the Hawkes ablation were fixed after seeing early results. No analysis was pre-registered.

9 Conclusion

We asked whether a market simulator that looks realistic gives reliable execution decisions. Without real order-level data we could not ask it of real markets. Asked of markets whose rules we know,

the answer in our experiment was no: the best-fitting simulator recommended a policy more than 20 bps worse than the best available, a cruder fit did better, a pretrained generative model could not be stabilised at all, and none improved on always choosing TWAP. The reason is that fit to unperturbed data leaves the response to intervention undetermined. Simulator validation for execution should therefore include what realism scores omit: stability without intervention, behaviour under interventions of the size to be studied, and comparison with simulator-free baselines.

References

Robert Almgren and Neil Chriss. Optimal execution of portfolio transactions. *Journal of Risk*, 3 (2):5–39, 2001.

Selim Amrouni, Aymeric Moulin, Jared Vann, Svitlana Vyetrenko, Tucker Balch, and Manuela Veloso. ABIDES-Gym: Gym environments for multi-agent discrete event simulation and application to financial markets. In *Proceedings of the Second ACM International Conference on AI in Finance*, 2021.

Emmanuel Bacry, Iacopo Mastromatteo, and Jean-François Muzy. Hawkes processes in finance. *Market Microstructure and Liquidity*, 1(1):1550005, 2015.

Tucker Hybinette Balch, Mahmoud Mahfouz, Joshua Lockhart, Maria Hybinette, and David Byrd. How to evaluate trading strategies: Single agent market replay or multiple agent interactive simulation? *arXiv preprint arXiv:1906.12010*, 2019.

Leonardo Berti, Bardh Prenkaj, and Paola Velardi. TRADES: Generating realistic market simulations with diffusion models. *arXiv preprint arXiv:2502.07071*, 2025.

David Byrd, Maria Hybinette, and Tucker Hybinette Balch. ABIDES: Towards high-fidelity market simulation for AI research. *arXiv preprint arXiv:1904.12066*, 2019.

Rama Cont, Sasha Stoikov, and Rishi Talreja. A stochastic model for order book dynamics. *Operations Research*, 58(3):549–563, 2010.

Weibing Huang, Charles-Albert Lehalle, and Mathieu Rosenbaum. Simulating and analyzing order book data: The queue-reactive model. *Journal of the American Statistical Association*, 110(509): 107–122, 2015.

Junjie Li, Yang Liu, Weiqing Liu, Shikai Fang, Lewen Wang, Chang Xu, and Jiang Bian. MarS: a financial market simulation engine powered by generative foundation model. In *International Conference on Learning Representations*, 2025.

Peer Nagy, Sascha Frey, Kang Li, Bidipta Sarkar, Svitlana Vyetrenko, Stefan Zohren, Ani Calinescu, and Jakob Foerster. LOB-Bench: Benchmarking generative AI for finance – an application to limit order book data. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, 2025.

Yosihiko Ogata. On Lewis' simulation method for point processes. *IEEE Transactions on Information Theory*, 27(1):23–31, 1981.

Svitlana Vyetrenko, David Byrd, Nick Petosa, Mahmoud Mahfouz, Danial Dervovic, Manuela Veloso, and Tucker Balch. Get real: Realism metrics for robust limit order book market simulations. In *Proceedings of the First ACM International Conference on AI in Finance*, 2020.

Yanzhi Zhang, Yu Ma, Yilin Cheng, Jian Li, and Yitong Duan. M3: A state-event generative foundation model for market microstructure dynamics. *arXiv preprint arXiv:2608.19227*, 2026.

A Reference worlds

Both mechanisms run on the same price–time-priority matching engine as the Hawkes simulator, with a 10 bps tick and lots of 100 shares, and are simulated by thinning state-independent candidate streams.

R1, queue-reactive. Order flow has no memory of past events; every intensity is a function of the current queues (Huang et al., 2015). Joining the best quote becomes less likely as that queue grows; limit orders arrive behind the best at a constant rate; quotes are improved only when the spread is at least two ticks; cancellations are proportional to resting depth; and a market order is more likely when the queue it would hit is short relative to the opposite best queue. A large sell thins the bid, which attracts both more selling and new bids. R1a is deep with a weak imbalance effect; R1b is thinner with a strong one.

R2, behavioural agent mix with a hidden regime. Noise traders submit limit orders, market orders and cancellations at constant rates. Fundamentalists send market orders towards a latent fundamental value that follows a random walk, with intensity growing as the 1.5 power of the mispricing. Trend followers send market orders in the direction of the recent return and are several times more active in a hidden "trending" regime that switches as a two-state Markov chain. Liquidity providers cancel quotes on a side that has just absorbed one-sided market volume. The fundamental and the regime are exogenous. R2a has strong fundamentalists; R2b has weak fundamentalists, stronger trend followers and faster withdrawal of liquidity.

What a candidate simulator sees. Six unperturbed 30-minute episodes per world: the event log (time, one of six event types, size, price level, price, and the spread and ten-level depths just before the event) and a 5-second grid of quotes and depth. It never sees the rules, the parameters, the fundamental value or the regime.

B Fitting the candidate simulators

Hawkes. Baselines and the 6×6 branching matrix are estimated by expectation–maximisation for each decay rate on a grid from 0.02 to 2 per second, keeping the rate with the highest likelihood. The simulator's baselines are state-dependent (cancellations scale with depth, market orders with the inverse spread), and the EM uses the same factors evaluated on the logged pre-event book state. Size and price-level marks are geometric, fitted by maximum likelihood; the probability of improving the quote is the share of limit orders placed inside the spread among those submitted when the spread allowed it. On data generated by the simulator itself, the fit recovers the decay exactly and the branching matrix within 0.04. If the fitted branching ratio reaches 0.98 the matrix is rescaled to 0.95; this was not needed in any world.

ABIDES. There is no likelihood. The mean price is set so that one cent equals one reference tick, and eighteen configurations (60, 102 or 180 value traders; none or two market makers; three levels of fundamental volatility) are each simulated without intervention for six seeds. The configuration

with the smallest realism error against statistics of twenty further unperturbed reference runs is kept.

MarS. MarS cannot be fitted. It was conditioned instead: the first thirty minutes of a MarS run replayed a reference episode's orders into MarS's exchange, after which the pretrained model generated all flow.

C Variance reduction in the reference worlds

Implementation shortfall splits exactly into the impact cost against the twin and a timing term, the twin's own drift weighted by when the policy traded. For a fixed schedule the timing term has mean zero in the reference worlds, because their rules are symmetric between buying and selling. The timing terms of TWAP, the $b = 4$ front-loaded schedule and the back-loaded schedule are therefore control variates with known mean: each policy's shortfall is regressed on them across seeds and the fitted part subtracted. This leaves every policy's expected cost unchanged, including the adaptive ones, and reduces the per-seed standard deviation by between a third and two-thirds where it is large (Table 5). The adjustment is applied only in the reference worlds; the candidate simulators' own estimates are plain means, as a user of those simulators would compute them.

D Additional pilot tables

Table 10: Pilot: implementation shortfall in bps (mean ± 95% half-width) for a parent order of 5% of background volume. Bold: lowest mean in the column. Costs are not comparable across columns: the worlds have different price levels and tick sizes.

Policy	A1	A5	H1	H2	M10	M2
P0 TWAP	-0.1 ±1.0	-0.2 ±0.8	13.7 ±7.7	24.8 ±23.4	905.1 ±1242.8	5.4 ±45.7
P1 Immediate	0.8 ±0.2	0.5 ±0.1	17.2 ±1.8	36.0 ±10.2	**20.0 ±14.1**	20.6 ±17.0
P2 Front-loaded (b=2)	-0.1 ±0.6	0.0 ±0.6	14.6 ±7.5	29.9 ±19.9	102.3 ±83.6	7.4 ±14.9
P2 Front-loaded (b=4)	0.1 ±0.5	0.1 ±0.4	15.2 ±5.9	31.8 ±15.4	35.0 ±26.2	16.2 ±25.9
P3 Back-loaded	**-0.4 ±1.1**	-0.4 ±1.0	**9.8 ±10.8**	21.1 ±27.8	34.1 ±12.6	27.9 ±31.5
P4 POV (1×)	0.0 ±0.9	-0.3 ±0.9	15.4 ±8.2	22.6 ±26.5	803.4 ±1016.9	26.8 ±27.7
P4 POV (2×)	-0.1 ±0.6	-0.0 ±0.6	16.3 ±6.3	37.4 ±19.8	31.6 ±32.3	9.4 ±34.5
P5 Liquidity-aware	-0.2 ±0.8	-0.2 ±0.9	12.2 ±8.9	**13.1 ±27.3**	881.3 ±1132.7	**-35.3 ±57.2**
P6 Signal (momentum)	0.2 ±0.9	0.4 ±0.9	13.6 ±8.2	20.7 ±26.4	68.1 ±77.1	4.2 ±22.0
P6 Signal (reversal)	-0.3 ±0.9	**-0.6 ±0.9**	14.5 ±8.6	13.5 ±26.6	406.0 ±574.0	13.9 ±29.4
P7 Risk-aware (0.5)	0.6 ±0.5	0.4 ±0.3	15.7 ±5.4	25.5 ±14.7	57.1 ±46.3	19.7 ±27.3
P7 Risk-aware (2)	0.8 ±0.4	0.4 ±0.3	16.7 ±4.5	33.8 ±13.9	35.6 ±18.5	6.8 ±6.4
Seeds	40	40	60	57	5	10
Half-spread (bps)	0.1	0.1	5.8	10.1	7.3	10.8

Table 11: Pilot: cross-simulator regret. Rows: world in which the policy was selected (first half of its seeds). Columns: world in which it is evaluated (second half). Entries are regret relative to the best candidate in the evaluation world, in units of that world's half-spread. †paired 95% interval above zero. A diagonal entry above zero means the policy selected on training seeds was not the best on test seeds.

Optimised in	Chosen policy	A1	A5	H1	H2	M10	M2
Parent order = 5% of background volume							
A1	Back	1.5	4.5†	0.0	0.2	2.0	9.0†
A5	Reversal	0.0	0	0.4	0.7	71.9	0.7
H1	Back	1.5	4.5†	0	0.2	2.0	9.0†
H2	Reversal	0.0	0.0	0.4	0.7	71.9	0.7
M10	Risk 2	16.3†	16.2†	1.1	2.5	4.4†	7.3
M2	Momentum	7.9†	15.9†	0.3	1.3	1.6	3.0†
Parent order = 15% of background volume							
A1	Front b=2	3.9	8.2†	1.9†	1.2	15.2	2.3
A5	Reversal	0.0	0	0.6	3.5†	82.6	5.0
H1	Risk 2	33.6†	20.0†	0.3	0.0	9.9†	4.0
H2	Risk 2	33.6†	20.0†	0.3	0	9.9†	4.0
M10	Momentum	5.2	14.4†	0.4	6.2†	26.9	2.3
M2	Reversal	0.0	0.0	0.6	3.5†	82.6	5.0

E Implementation notes

MarS in-process. The model is loaded with the repository's own class and sampled with its own procedure (softmax at temperature 1, one multinomial draw); only the HTTP round trip to a Ray Serve deployment is removed. The repository's Ray import is satisfied by a stub. Child orders are marketable limit orders priced at the deepest visible level they need and never larger than the ten visible levels, since the engine has no market-order type. Rollouts are limited to 420 seconds of wall-clock time for the 50-minute prefix and 150 seconds for each 12-minute branch.

ABIDES. The order-size model depends on an old release of the `pomegranate` library; a small module sampling the same mixture replaces it. To place ABIDES at a lower price, the fundamental volatility and the two shock parameters of the RMSC04 configuration are scaled with the price level; at the default price the original results are reproduced bit-for-bit.

Tests. Thirteen unit tests cover the sign of costs for buys and sells, the charge for unfilled shares, clipping, lot rounding and forced completion, completion of the parent order by every candidate policy, invariance of each policy's decisions to later observations, price–time priority in the matching engine, subcriticality of the Hawkes regimes, and identity of twin and intervention before the task.